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CSE360: Computer Interfacing

Lab Project Report

Smart Portable Human Safety Device for Industrial Worker

by

[Group-08]

[Section-07]

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Abstract

The objective of this project is to develop and introduce a smart, compact safety gadget to prevent industrial accidents among the workforce in Bangladesh by proactively protecting them against common industrial accidents. The system is designed to use a microcontroller with an integrated set of sensors, such as a gas sensor (MQ-2), an air quality sensor (MQ-135), a temperature-humidity sensor (DHT11), a pulse sensor, and an accelerometer (MPU6050) to detect falls. The technique is to constantly monitor the surroundings and physiological activity of the worker in real-time. When a threshold has been crossed (e.g., a gas leakage, abnormal temperature, irregular heart rate, or a fall), the device sends instant notifications by vibration through the buzzer and LED. Importantly, it uses Bluetooth as an emergency signal transporter to the paired smartphone to enable fast rescue missions. Moreover, the system was provided with an RFID-based access control that includes a security measure to allow only authorized personnel to use the device and an LCD to present essential data in order to monitor this or that data on the spot. These findings ensure the effectiveness of the system as an efficient, one-stop solution to improving occupational safety, with an important possible effect on minimizing accidents and ensuring sustainable worker welfare in dangerous industrial environments.

Keywords

[Safety device, Industrial IoT monitoring, RFID access control, Bluetooth emergency alert]

1. Introduction

1.1 Problem and Motivation

The industrial sector in Bangladesh is a pillar of the economy, yet the safety challenge is high. Workers in garment, manufacturing, and chemical plants are frequently involved in life-threatening risks, including gas leakages that may be invisible, poor air quality, excessively high temperatures, and the threat of falls or accidents related to health conditions induced by fatigue. The previous approaches to safety tend to be reactive, fixed, insufficient, and fail to provide personal real-time safety. These risks are increased by the lack of timely alert mechanisms and effective emergency response systems, which contribute to accidents in the future, chronic health conditions, and deaths. This alarming gap in occupational safety demands a new, proactive, and comprehensive solution that considers the needs of the individual worker.

1.2 Project Objectives

The main goal of the proposed project is to create and deploy a smarter, more compact, and wearable safety device that will serve as an all-around protection for an industrial operator. The specific goals are:

- To constantly observe the surroundings of the worker for the presence of hazardous conditions such as exposure to toxic gases, poor air quality, and extreme temperature/humidity.
- To control the physiological health of the worker using heart rate sensing and automatic fall warning.

- To send instant multisensory (audio, visual) local alerts to the worker when any danger is detected.
- To facilitate fast emergency response by sending alert messages and real-time location information to distant supervisors through wireless communication protocols.
- To provide responsibility and permitted use by an RFID-based access control system.

1.3 Relevance to Computer Interfacing

This project plays a vital role in learning computer interfacing. In our theory lectures, we learned how to interface a system so that humans can communicate with it. In this project, we are using an Arduino UNO that needs to be interfaced so that it can be human-friendly. We can use the sensors to communicate with the Arduino, and we can also use the buttons to send requests to the system. By default, the system cannot take these commands without computer interfacing. Finally, by using computer interfacing in our Arduino, we are programming our project to communicate and serve its purpose.

2. Related Work/ Inspiration

There are very few works using safety vests. There is a [project](#) that can only detect metals in a safety vest. But in industrial situations, it lacks major features that can save many lives.

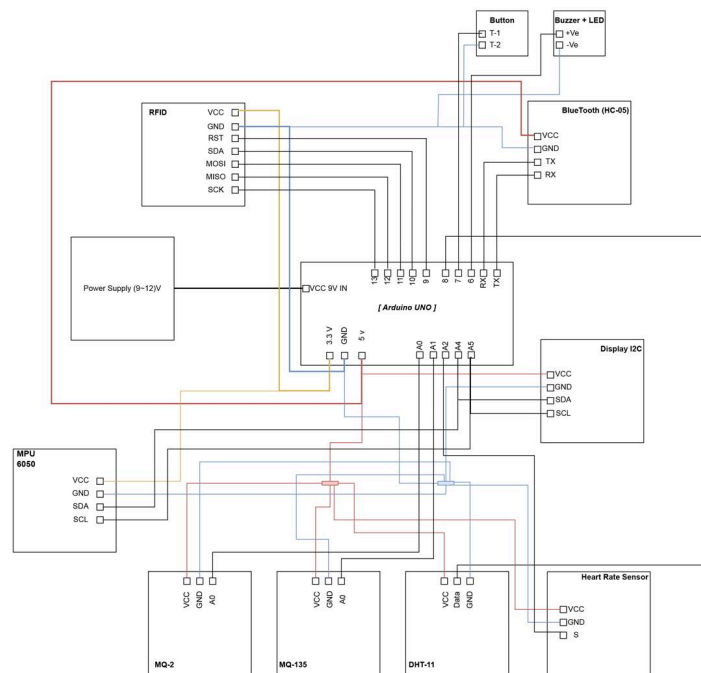
The key distinctions of our project are:

1. It is multifunctional and practically designed.
2. It can show various information in a display (temp, heart rate, and air quality) that other vests lack currently
3. Fall detection in industrial situations.

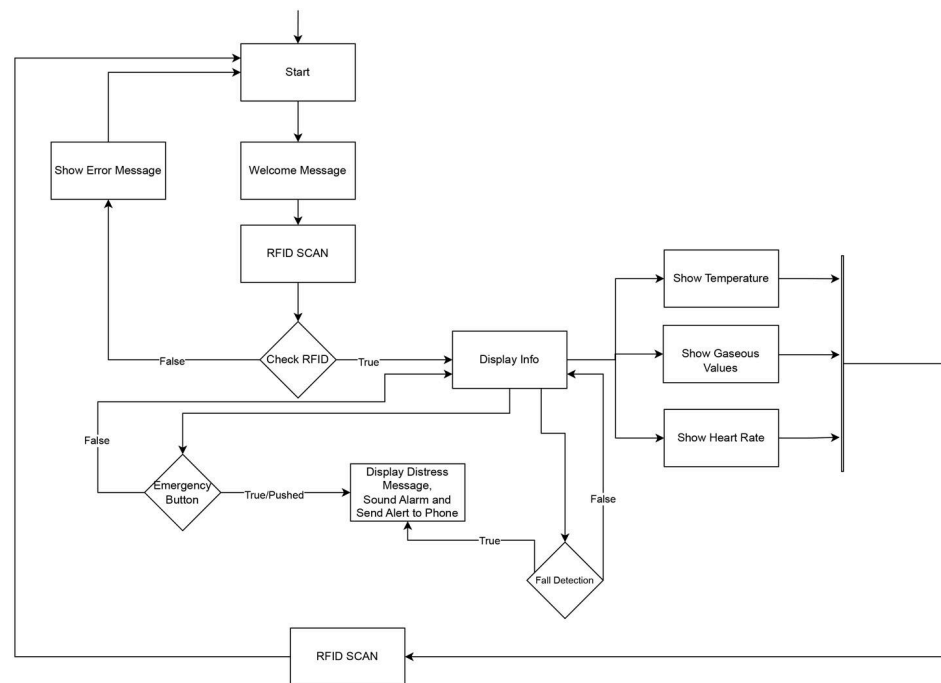
4. Bluetooth Connectivity with the phone.
5. RFID-Based Authentication Check.

3. Technical Approach

Block Diagram:



Flow chart:



Components Used:

1. **MQ-2 Gas Sensor** - Mainly, we are using it for gas detection. Also, it detects gases like smoke, hydrogen, and methane.
2. **MQ-135 Sensor** - It detects air quality (CO₂, alcohol, NH₃, smoke, etc).
3. **LEDs** - Used for indicators or warnings.
4. **BUZZERS 5V** - Used for alarms.
5. **MQ-6050** - It's used for fall or crash detection.
6. **Heart Rate Pulse Sensor** - It measures heartbeat by detecting the blood flow.
7. **RCS22 RFID Card Reader Module** - It reads RFID tags or cards for authentication purposes.

8. Micro Card Module - Used for reading/writing data to an SSD Card by Arduino.

9. DS3231 RTC Module -It keeps accurate data and time, also a Real-Time Clock module.

10. Wires (Male & Female) -Jumper wires to make connections on breadboard/Arduino.

11. Bread Board - Board board used to connect component prototyping without soldering.

12. Arduino UNO - Primary microcontroller board that manages all sensors, inputs, and outputs.

13. Software (Arduino IDE) -The environment used to create and upload code to Arduino.

Cost Breakdown:

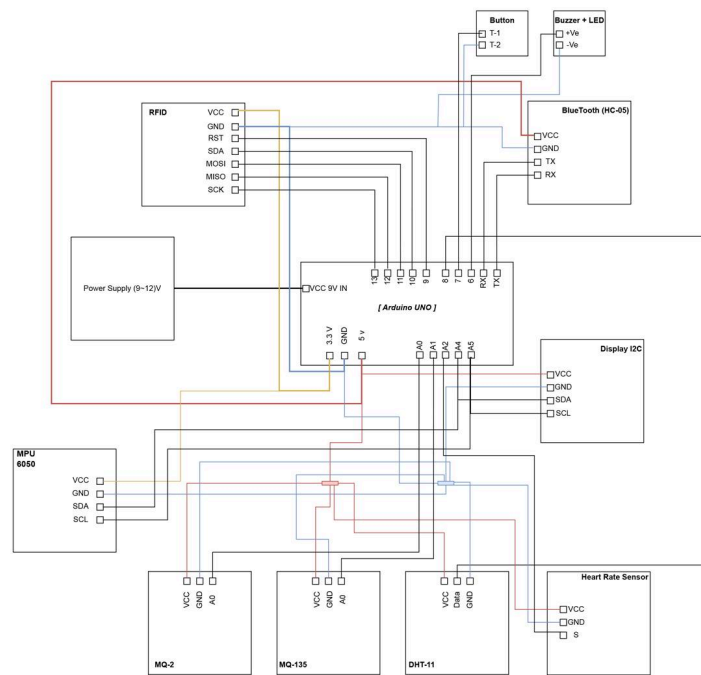
Items	Quantity	Amount
MQ-2 Gas Sensor	1	140
MQ-135 Sensor	1	150
LEDs	1	5
BUZZERS 5V	1	10
MQ-6050	1	220
Heart Rate Pulse Sensor	1	260
Vest	1	250
RCS22 RFID Card Reader Module	1	180

MicroSD Card Module	1	60
DS3231 RTC Module	1	220
Wires (Male & Female)	50	50
Arduino UNO	1	650
Soldering Iron	1	220
Tape	1	70
Wires for connection	9.1 M	200
TOTAL		2,685 BDT

Communication Protocol:

In our project, we are using communication protocols such as I2C (Display, MPU6050), SPI for RFID, UART for Bluetooth communication for the rest of the sensors, we are taking analogue inputs using a single data line. By using this combination, it ensures responsiveness, and they are adaptable with the Arduino libraries, ensuring a reliable system integration.

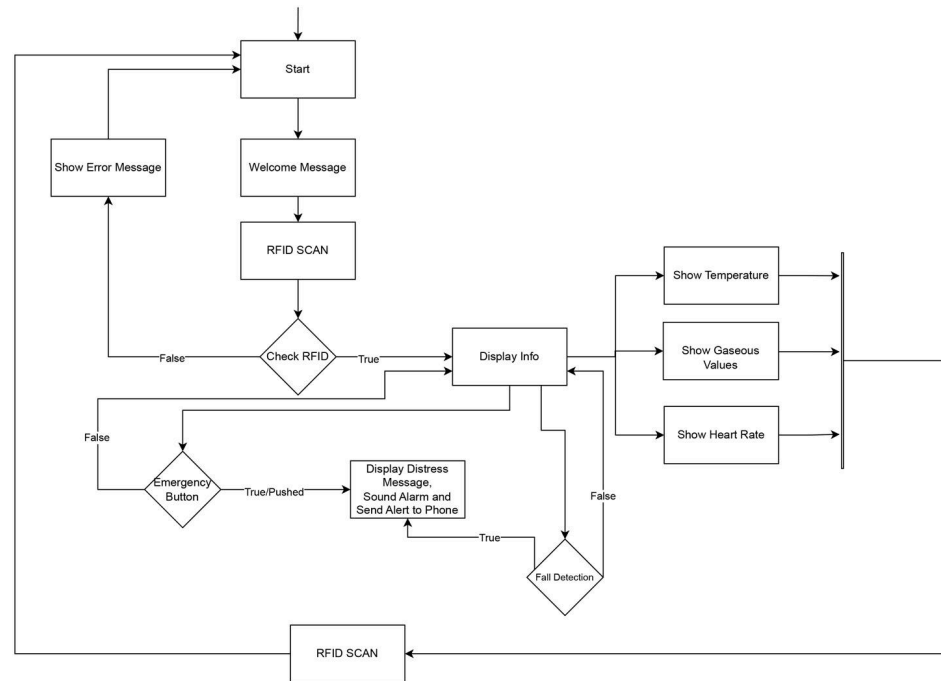
Circuit/Schematic:



Implementation:

First, the person turns on the safety vest. The vest is disengaged due to authentication verifications. He/She swipes the RFID card, and the vest verifies the credentials. If the values match, it will start working and display all necessary data; if the values do not match, it will not allow the data to be displayed and show an error message. After the authentication, the user can view the basic data, such as toxic gas values, temperature, and heart rate values when the finger is placed on the heart sensor. If the person is in danger, then he/she can press the emergency SOS button that will play an alarm sound and notify the administrator using Bluetooth protocol. If he/she falls from a height or a major movement is detected, the vest will play an alarm sound and again notify through Bluetooth for a fall/crash detection. If the toxic gas values exceed a normal

range, the vest will also play an alert notifying the user to retreat to safety. Finally, after work, if he/she swipes the RFID, the vest will shut down and return to the welcome screen.



Challenges:

1. Sensor communication issues were faced with the temperature sensor and later fixed by soldering the wires and updating the code.
2. Power supply inability was found, as very few batteries/power banks can supply a steady voltage of 9V for a long time. Later, that was fixed by using a router UPS.
3. Designing issues faced as the vest is made of thin material, later fixed by using paper, glue, and tape to attach the cables.

4. Bluetooth communication issues are faced as there are no apps to integrate the notification, alarm from the phone. Later, a Bluetooth serial monitor app is used.
5. Major RFID issues are seen as the board can not hold a steady supply of voltage for this many components. As a result, two RFID boards are destroyed in the assembly process and are later replaced with another module.
6. SD card irrigation failed as the code doesn't support the functionalities of Arduino UNO, and the RAM usage becomes (80~85)%. As a result, the full system hangs and the functionalities freeze. Due to technological barriers, this problem could not be overcome.
7. While connecting the components, a team member almost fainted due to electrical shock from the soldering iron, later solved by wearing ground-insulating slippers.

4. Sustainability & Impact

Sustainability: This device employs energy-efficient sensors and low-powered microcontrollers in its design. Rechargeable batteries are used to avoid frequent replacements. Eco-friendly and recyclable materials are used for the vest to reduce environmental impact. The system ensures sustainability and protects the workers by merging energy-saving technology with reusability.

Impact: This safety device improves worker protection, reduces accidents, and supports quick emergency response in industries. It helps create a safer working environment in Bangladesh, protecting workers' health in industries.

Future Work: Future improvements can include integrating IoT and cloud systems for remote monitoring, GPS tracking for outdoor industries, and AI-based predictive analysis. The device can also be scaled for use in construction and other hazardous sectors. Lighter materials and advanced sensors may further enhance comfort and reliability.

Limitations: Some hardware limitations include sensor calibration issues, limited battery backup, and possible inaccuracies in extreme environments. Storing data in a microSD card failed due to the storage limitation of Arduino Uno, because this functionality uses 80 to 85% storage of the Arduino, which exceeds the capacity of the Arduino. Additionally, we have not created a mobile application to directly send the danger notification to the phone via Bluetooth communication, which is why we have to use the Serial monitor app to observe real-time notifications. Despite these challenges, the project shows strong potential for real-world application.

5. Results & Discussion

By using our project, we can estimate that the accuracy is very high, and the detection is almost instant. And by using our device, the usability of the vest increases significantly. To elaborate the gas sensor's accuracy resulted in a $\pm 90\%$, also fall detection accuracy is higher due to testing purposes, as we cannot check fall detection through the system, but it is accurate above $\pm 80\%$, Bluetooth and RFID accuracy was the highest instant response from the device. To summarize the details in a chart below:

Feature	Sensor/Module Used	Accuracy
Gas Leak Detection	MQ-2	90%
Air Quality Monitoring	MQ-135	
Fall Detection	MPU6050	80%

Temp & Humidity Monitoring	DHT11	$\pm 2^{\circ}\text{C}$
SOS & Bluetooth Alerts	Push button HC-5	Triggered
Pulse Monitoring	Pulse Sensor	95%
RFID Access Control	RFID Module	100%

The project showed its primary objectives of ensuring worker safety through real-time monitoring and a rapid alert system. Gas and air quality sensors were also found to be reliable in monitoring the deadly conditions, and this is vital in an industrial setting. Fall detection was good, but false positives occurred in cases where there was rapid movement. The pulse and temperature monitoring offered reliable and stable information that reinforced the health monitoring of the device. The panic button and the Bluetooth connection were used to make sure that SOS alerts were sent promptly, but the range was no more than 10 meters. Access control using RFID measures was effective in limiting the use without authorization, which is in line with the industrial safety requirements.

In general, the system had a solid potential of enhancing industrial safety in Bangladesh.. These findings attest to the fact that the device achieves the desired goals of anticipatory risk identification, emergency reaction, and worker welfare surveillance. But other issues, like sensor calibration, wireless range, and power optimization, are some fields that could be improved.

6. Conclusion

The project of the Smart Portable Human Safety Device to be used by the Industrial Workers is a successful example of how computer interfacing and embedded systems can contribute greatly to the safety in the workplace. The system brings various sensors, that is, gas, air quality, temperature-humidity, pulse, and accelerometer, and RFID authentication and Bluetooth

communication, which gives the device the ability to monitor the environment in real-time and send immediate alerts in dangerous situations.

Key achievements include:

- Stable multi-sensor combination to identify environmental and physiological hazards.
- Immediate alert mechanisms via buzzer, LEDs, and Bluetooth for faster emergency response.
- RFID authentication to provide authorized use and data security.

The project fills in important gaps in industrial safety with an active and simple solution. Its sustainable construction through energy-saving materials and recyclable resources indicates its promise of long-term industrial application. This device, with further additions of IoT connectivity, GPS tracking, and AI-driven predictive analysis, can be included in the list of reducing work-related accidents and protecting the welfare of workers in high-risk workplaces.

7. Contribution

Name	ID	Role(s) / Responsibilities	Specific Contributions

Faishal Monir	22101235	Assembling the system in the vest, implementing the feature of fall detection, and coding parts of the project.	● Assembled the vest ● Soldered the final connections in the vest ● Handled the coding parts of fall detection (MPU6050)
Rifat Mahmud Tamim	21201787	Helping with the assembly of the breadboard of the project. Implementing Bluetooth and RFID in the project.	● Assembled the RFID in the breadboard ● Tested the Bluetooth serial communication using Arduino ● Handled the coding part of Bluetooth and RFID
Umma Salma Mim	22101870	Helping in assembly with the vest, integrating the smoke sensors and pulse sensor for the vest, and overall supervising the full project.	● Helped with the Vest assembly ● Tested the smoke detectors for tolerance ● Implemented the Smoke and pulse sensors and their codes (MQ2, MQ135, and Amped Sensor)

Rajin Ibna Rajuanur Rahman	22101717	Logistic support with hardware implementations (Main feature developed: Panic button implementation) and testing, code debugging, and correction.	● Handled the supply of all components ● Testing other code for major errors and solving those issues ● Implementing the panic button feature with code and the wireless Bluetooth message using Tamim's code.
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<https://www.arduino.cc/en/Guide/ArduinoUno/>
3. *Kartik9250/Fall_detection: The ESP32 Fall Detection Device is a wearable gadget that uses an MPU6050 sensor to detect falls based on acceleration jerk. It triggers a buzzer during a fall and includes an interrupt button to stop it. With a built-in WiFi manager, it connects to networks and sends SOS SMS messages to emergency contacts for help.* (n.d.). GitHub.
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